

Effect of Hierachy on the Mechanical Performance of Honeycomb Structures

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This study compares the mechanical properties of a honeycomb structure versus its first-order hierarchy variation. Through simulation, the two types of specimens were subjected to in-plane tension and compression, and out-of-plane tension to compare the effective stiffness, maximum principal stress, and maximum strain energy. For both specimens, overall dimensions and carbon fiber-Onyx composite material makeup were kept fixed to isolate the results of manipulating the honeycomb geometry. Results from these simulations along with other papers show that first-order hierarchal honeycomb structure has higher stiffness in in-plane compression than the default honeycomb geometry, while performing worse for in-plane tension.

I. INTRODUCTION

A. Background

Honeycomb composite structures, which are characterized by a periodic arrangement of multi-cellular geometries, are among the most widely studied cellular materials in modern engineering. They are valued for their high strength-to-weight ratio, energy absorption capacity, and structural efficiency. The unique periodic lattice yields a structural system that minimizes mass while retaining mechanical performance. As a result, honeycomb cores are widely used in aerospace sandwich panels, automotive crash structures, packaging and biomedical applications, where the combination of low mass and high stiffness is critical [1].

While seemingly simple, honeycomb structures go beyond the traditional hexagonal shape through various cell shapes and in-plane topological arrangements. The mechanical behavior of a honeycomb is dictated more by the geometry of its unit cell rather than the bulk properties of its constituent material. Key parameters, including cell wall length, thickness, height, and the angle between walls, govern the effective stiffness, strength, and failure modes of the structure under different loading conditions. Research from [1] has established scaling laws that relate geometric parameters and relative density of the cellular structure to its mechanical properties. These relationships underpin the analytical and computational studies of honeycomb design, and they form a useful baseline against which novel cellular geometries can be compared [2].

B. Hierarchical Honeycombs

As the mechanical behavior of conventional regular honeycombs is well-established, recent research has increasingly shifted towards hierarchical cellular structures, in which the walls of a primary honeycomb structure themselves are composed of smaller cellular features. This concept is inspired by materials like wood, bone, and pomelo peel, where multi-scale hierarchical organization contributes to remarkable combinations of stiffness, toughness and low density [3].

The work of Ajdari et.al [4] introduced the concept of geometric hierarchy by replacing each three-edge vertex of a standard hexagonal lattice with a smaller sub-hexagon, generating what is known as a first-order hierarchical structure. Their work shows that at constant relative density, first order and second order honeycombs can exhibit higher amounts of specific stiffness compared to a regular honeycomb. Subsequent studies have confirmed and built on these findings, showing improvements in load-bearing capacity, energy absorption, and stress redistribution at vertices [4].

However, it is critical to note that performance gains are not uniform across all loading modes. While substantial improvements in in-plane compression and crushing energy absorption are well-documented, the structural response under out-of-plane tension and complex bending loads remains nuanced and dependent on precise geometric parametrization. For instance, improvements in in-plane compression and crushing energy absorption are well documented [5], while behavior under out-of-plane and bending loads is more nuanced and depends strongly on geometric parameters [6].

The rapid advancement of additive manufacturing has further expanded the practical relevance of these geometries. Fused deposition modelling (FDM) using fiber-reinforced thermoplastics allows for complex cellular geometries to be fabricated with composite materials that offer better mechanical properties [7]. Markforged Onyx, a polyamide matrix reinforced with chopped carbon fibers, is one such material. It offers stiffness and strength 1.4 times that of unfilled ABS while maintaining good dimensional stability and surface finish, while reinforcement with continuous fiber boasts even greater

strength [8]. These attributes make Onyx a suitable candidate for the fabrication and study of hierarchical cellular structures.

C. Scope and Contribution

Despite extensive research on individual aspects of honeycomb mechanics, comparative studies that evaluate the same geometric variations across multiple loading modes remain limited. In particular, the trade-offs between regular and first-order hierarchical configurations under in-plane tension, in-plane compression and out-of-plane tension are not always clearly delineated, and the optimal geometry is known to depend strongly on the loading scenario [9].

This paper presents a simulation-based analysis of how implementing a first-order hierarchy of honeycombs influences the mechanical performance of honeycomb structures fabricated from Markforged Onyx. Geometries are modeled in SolidWorks and AutoCAD and analyzed in ANSYS under the three loading conditions noted above.

II. OBJECTIVES

The objective of this paper is to map how hierarchy of honeycomb structures affect performance across loading modes and to identify where they offer distinct advantages over regular honeycomb structures. By comparing regular and hierarchical configurations, this paper clarifies which architectures are best suited for varying weight constraints. Rather than identifying a single optimal design, these findings will hopefully support more informed honeycomb design choices for applications where loading conditions and weight constraints vary.

The specific objectives of this paper are as follows:

1. To develop parametric CAD models of regular and first order hierarchical hexagonal honeycomb structures, with cell wall length, thickness and height remaining constant.
2. To simulate the mechanical response of regular and first order configuration in ANSYS under four distinct loading scenarios: in-plane tension, in-plane compression, and out-of-plane tension.
3. To extract and compare key performance metrics, including effective stiffness, peak load, and specific properties normalized by mass across parameters.
4. To highlight design regimes in which one configuration offers a clear advantage over the other, providing practical guidance for honeycomb design under loading conditions.

The working hypothesis is that first-order hierarchical honeycombs will outperform regular honeycombs in in-plane compression at comparable mass, consistent with prior findings on stress redistribution at vertices. It is expected that this advantage will diminish or reverse under out-of-plane tension, where loading is carried more by the cell walls than through joint regions.

III. METHODS

This section describes the workflow used to model, simulate, and analyze the honeycomb structures. The study uses a combination of parametric CAD modeling in SolidWorks and finite element analysis (FEA) in ANSYS to evaluate the mechanical performance of each configuration under the three loading conditions.

A. Geometry and Parameterization

Two honeycomb configurations are considered in this study: a regular hexagonal honeycomb and a first-order hierarchical honeycomb. The first order hierarchical configuration is generated by replacing each three-edge vertex of the regular hexagonal lattice with a smaller hexagon. The hierarchical ratio is defined as:

$$\gamma_1 = \frac{b}{a}$$

In this formula, b is the smaller hexagon edge length, a is the parent hexagon edge length. This paper analyzes the ratio $\gamma_1 = 0.4$, where $b = 15$ and $a = 37.5$.

The geometric parameters of the honeycombs are summarized below:

1. **Cell wall length (a):** the edge length of the parent hexagon, fixed at $a = 37.5 \text{ mm}$ for the regular honeycomb. For first order honeycomb, the edge length of the smaller hexagon is fixed at $a = 15 \text{ mm}$.
2. **Cell wall thickness (t):** the thickness of the hexagonal walls. The minimum thickness is constrained to $t = 3.75 \text{ mm}$ which is constrained by the minimum printing thickness of fiber by the Markforged printer.
3. **Cell wall height (h):** held constant to $h = 25.4 \text{ mm}$ (1 inch) for all configurations to isolate the effects of in-plane geometry and hierarchy.

The relative density of each configuration is computed analytically and verified against the SolidWorks mass properties tool. Performance metrics are normalized by mass to allow for fair comparison between regular and hierarchical configurations of differing relative density.

The planar contact areas of the regular and first order honeycomb structures are found to be:

- **Regular:** 147153.503 mm^2
- **First order:** 131917.97 mm^2

The geometries of the regular and first order honeycomb structures are shown in Figures 1,2 and 3 below:

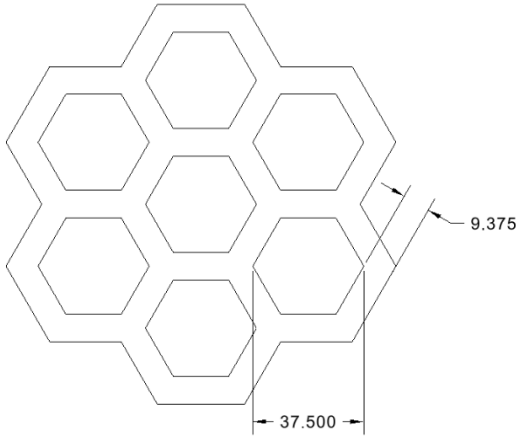


Fig. 1. Default regular honeycomb geometry with $a = 37.5\text{mm}$ and $t = 9.375\text{mm}$.

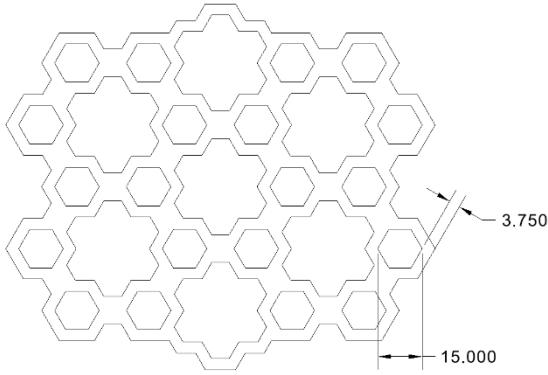


Fig. 2. First-order honeycomb geometry. Hexagons are 2.5x smaller than those of the default honeycomb geometry with $a = 15\text{ mm}$ and $t = 3.75\text{ mm}$.

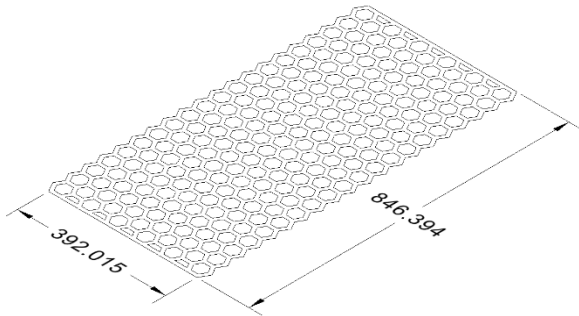


Fig. 3. Overall specimen dimensions for both geometries.

B. Material Model

The constituent material used in this study is the Markforged Onyx, a polyamide matrix reinforced with chopped microfibers, used in FDM processes. It is assumed that the chopped fibers are short and randomly oriented within the matrix, so the bulk material is treated effectively isotropic and linear elastic for the purposes of this study. The carbon fiber reinforcement is treated as an orthotropic material.

The material properties for Onyx used in the FEA model, obtained from the material datasheet and other simulation work [7] are:

- Young's Modulus (E): 2.4 GPa
- Poisson's ratio (ν): 0.39
- Density (ρ): 1200 kgm^{-3}
- Tensile strength: 36 MPa
- Flexural strength: 81 MPa

It is worth noting that the assumption of isotropy is acknowledged as a simplification, as real life printed Onyx maybe exhibit anisotropy due to layer-by-layer deposition.

The material properties for carbon fiber used in the FEA model, also obtained from the other simulation work [7] are:

- Young's Modulus (E_x): 60 GPa
- Young's Modulus (E_y, E_z): 8.5 GPa
- Poisson's ratio (ν_{yz}): 0.12
- Density (ρ): 1400 kgm^{-3}

C. CAD Workflow

All honeycomb geometries are modeled in SolidWorks using a parametric approach. A unit cell is constructed first, defined by the parameters a , t and h , and then patterned to form a representative volume of the honeycomb structure. For the first order hierarchical configuration, the smaller hexagonal substructure is added at each vertex of the parent hexagon and similarly patterned across the structure.

To minimize boundary effects in the FEA results, each model contains enough unit cells in the direction of loading to approximate bulk behavior. Models are exported in STEP format and imported into ANSYS for meshing and analysis.

D. Finite Element Analysis Set-Up

Simulations are performed in ANSYS Mechanical. After defining the material properties in Engineering Data, the imported geometry was reduced to the midplane surface assigned with a named selection.

A mesh convergence study is performed for one representative geometry under each loading mode. The mesh is progressively refined until the difference in output metric (effective modulus or peak load) between successive refinements is less than a tolerance of 5%. To achieve a

reasonable simulation time with accurate results, a mesh size of $0.01m$ is used across all simulations.

The surface model is then opened in ACP (Pre), where the Onyx and carbon fiber fabrics are defined. For better computational efficiency, we defined the 1-inch-thick part to be composed of 20 layers, thus treating each ply as 1.27 mm thick. Mimicking the standard Eiger slicing process, the first and last four layers were Onyx, while layers 4-8 and layers 12-16 were carbon fiber. For simplicity, carbon fiber was assumed to be unidirectional along the defined-rosette X-axis, while the Onyx layers were made alternating [45/-45]. The entire part is symmetrical.

The oriented selection set was defined, the plies and their orientations were set in the modeling group, and the solid model was defined. This model was imported into Static Structural.

E. Loading Cases

Four loading cases are simulated. The honeycomb coordinate system is defined such that X1 and X2 lie in the plane of the hexagonal pattern, and X3 is parallel to the cell wall height (out-of-plane direction).

1. In-plane tension

The honeycomb specimen is loaded in tension along the X2 direction. One in-plane face is fixed, and a uniform tensile force of $F_y = 100000N$ is applied to the opposite face in the X2 direction. Lateral surfaces are left free to allow for Poisson contraction.

In this loading mode, cell walls deform primarily by a combination of bending and stretching at inclined walls, producing effective stiffness that is lower than the out of plane case.

2. In-plane compression

The specimen is loaded in compression along the X2 direction. One in-plane face is fixed, and a uniform compressive force of $F_y = 100000N$ is applied to the opposite face in the X2 direction. Lateral faces are once again left free to allow for Poisson contraction.

In this loading mode, the cell walls deform primarily by bending, producing significantly lower effective stiffness than the out-of-plane case. Since the structure is regular hexagonal, the response in X2 direction is equivalent to X1 by symmetry.

3. Out-of-plane tension

The honeycomb specimen is loaded in tension along the X3 direction, parallel to the cell walls. The bottom face of the specimen is fixed in all directions, and a uniform force is applied in the X3 direction of $F_z = 100000N$.

In this configuration, the cell walls deform primarily by axial stretching, leading to high effective stiffness and strength relative to in-plane loading.

F. Output Metrics and Normalization

For the in-plane tension, in-plane compression and out-of-plane tension simulation, the following metrics are extracted:

- **Directional deformation (δ):** The strain in the direction of force (X3 for out-of-plane and X2 for in-plane) is determined.
- **Effective stiffness (E^*):** This is calculated from the deformation (δ) obtained above, where F = applied force on face, A = area of face, δ = directional deformation and L = length in direction of applied force.

$$E^* = \frac{F/A}{\delta/L} \quad (2)$$

- **Maximum principal stress ($\sigma_{1,max}$):** Failure modes are determined by maximum principal stresses; hence their magnitude and location are of interest. The location of $\sigma_{1,max}$ provides insight on where failure is likely to initiate and gives an idea on how well stress is distributed.
- **Maximum strain energy (U_{max}):** The total elastic strain energy stored in the structure is extracted from ANSYS. This metric provides a measure of the structure's capacity to absorb energy elastically before failure. It is informative when comparing specimens of similar mass, as hierarchy structures are expected to redistribute strain energy across more cell walls.
- **Mass of specimen:** This is generated from the volume and density values in SolidWorks for an estimation of the specimen mass.
- **Specific values:** Given that the lightweight application of honeycomb structures is its main strength, all the above metrics are reported in a mass-normalized form to allow for fair comparisons between configurations of different masses.

IV. RESULTS

Results from SolidWorks show that the masses of regular and hierarchy honeycomb to be:

TABLE I. EFFECTIVE MASS OF HONEYCOMB STRUCTURES

Honeycomb	Mass (g)
Regular	4485
First Order	4020

A. In-Plane Tension

This section describes the performance of regular and first order honeycomb structures under in-plane tension with similar boundary conditions and applied force.

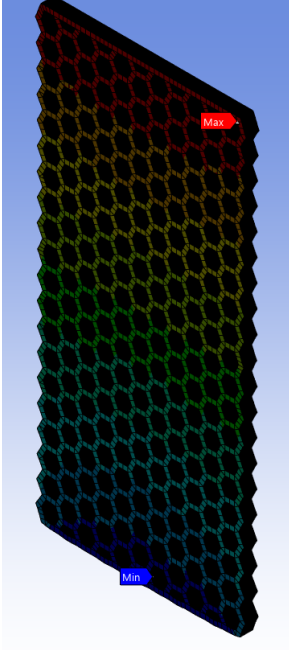


Fig. 4. Directional deformation for regular honeycomb.

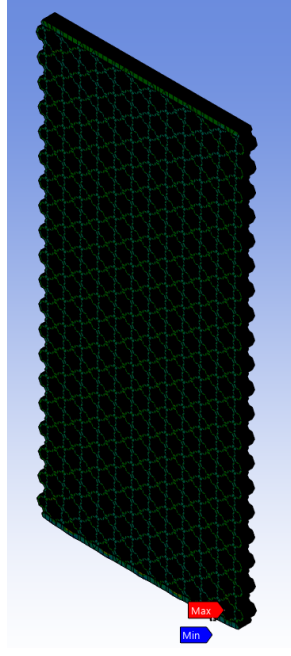


Fig. 5. Directional deformation for first order honeycomb.

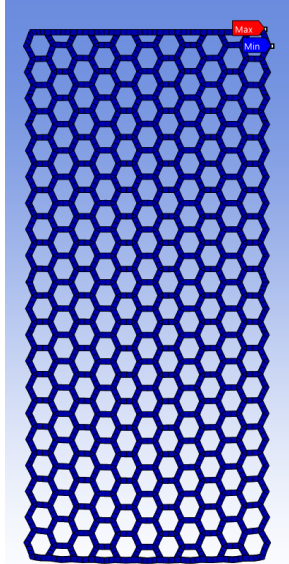


Fig. 6. Maximum principal stress for regular honeycomb.

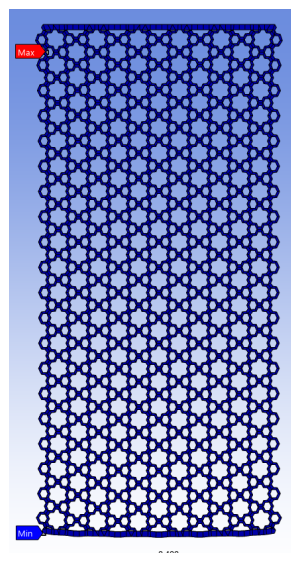


Fig. 7. Maximum principal stress for first order honeycomb.

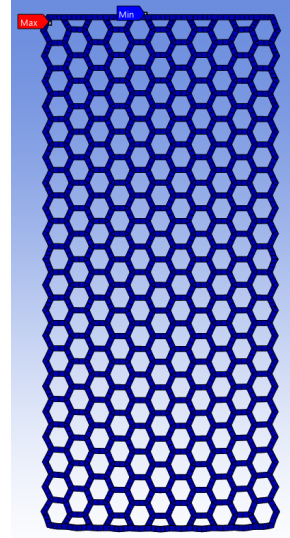


Fig. 8. Maximum strain energy for regular honeycomb.

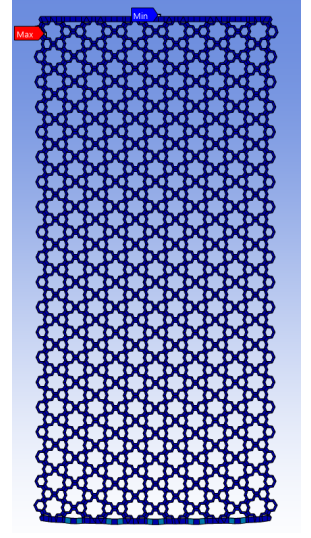


Fig. 9. Maximum strain energy for first order honeycomb.

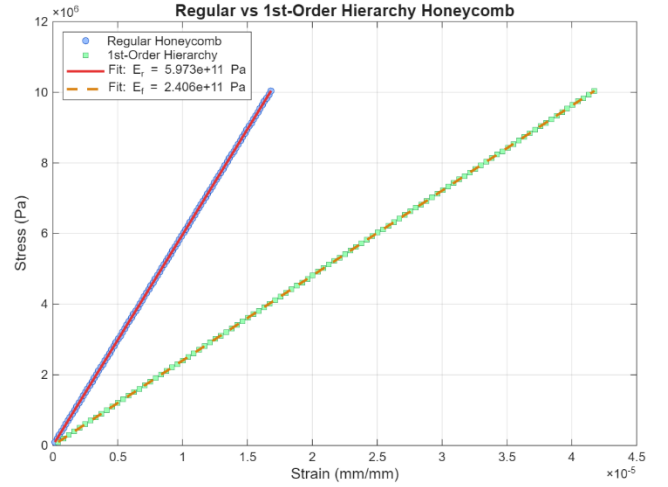


Fig. 10. Effective stiffness plots for regular and first order honeycomb.

TABLE II. IN-PLANE TENSION RESULTS

Honeycomb	Effective Stiffness, E^* (GPa)	Maximum Principal Stress, σ_{1max} (MPa)	Maximum Strain Energy, U_{max} (J)
Regular	597	331	3.1e-02
First Order	241	2780	2.0286

TABLE III. NORMALISED RESULTS

Honeycomb	Specific E^* (GPa/kg)	Specific σ_{1max} (MPa/kg)	Specific U_{max} (J/kg)
Regular	133.11	73.801	6.912e-03
First Order	59.95	691.542	0.505

B. In-Plane Compression

This section describes the performance of regular and first order honeycomb structures under in-plane compression with similar boundary conditions and applied force.

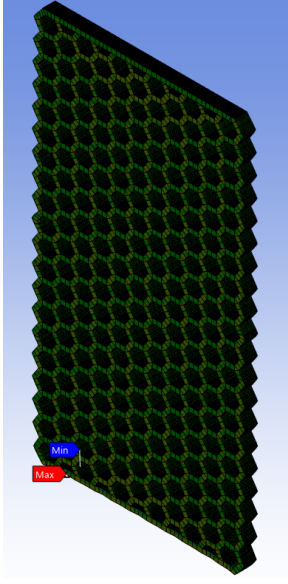


Fig. 11. Directional deformation of regular honeycomb.

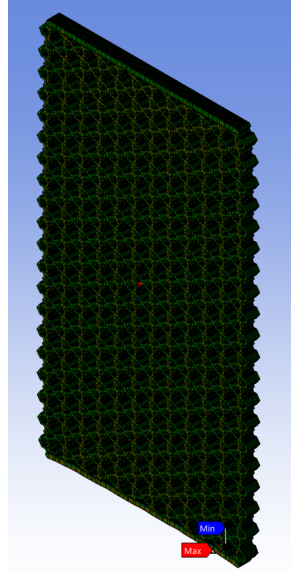


Fig. 12. Directional deformation of first order honeycomb.

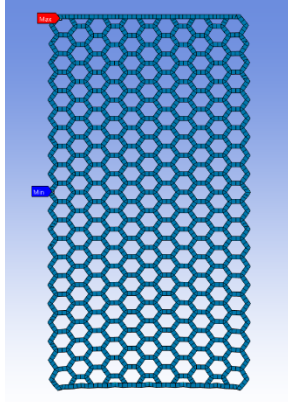


Fig. 13. Maximum principal stress for regular honeycomb.

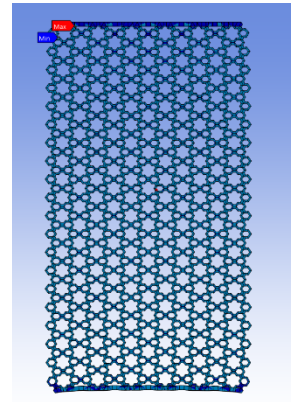


Fig. 14. Maximum principal stress for first order honeycomb.

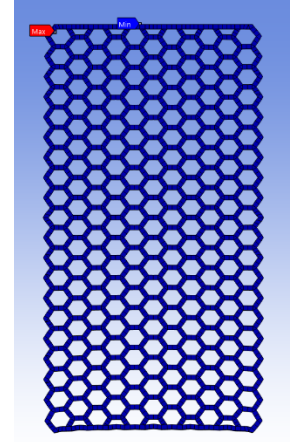


Fig. 15. Maximum strain energy of regular honeycomb.

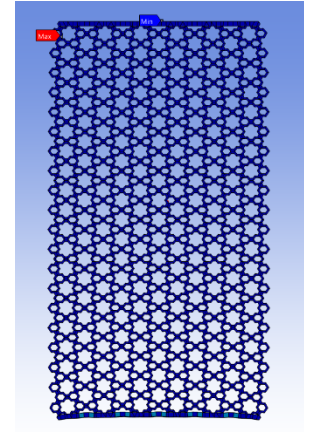


Fig. 16. Maximum strain energy for first order honeycomb.

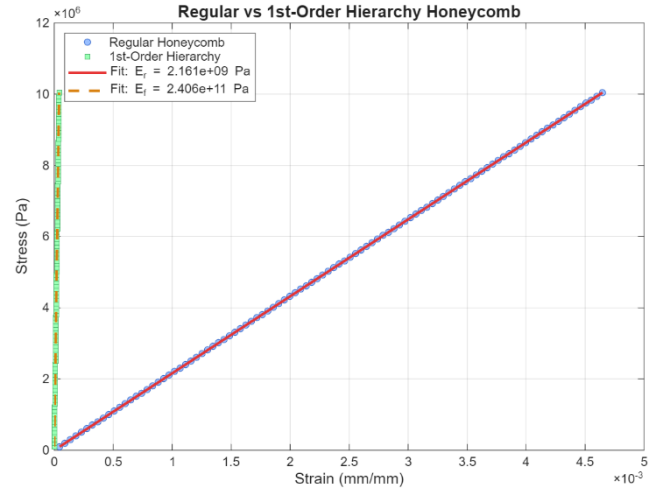


Fig. 17. Effective stiffness plots for regular and first order honeycomb.

TABLE IV. IN-PLANE COMPRESSION RESULTS

Honeycomb	Effective Stiffness, E^* (GPa)	Maximum Principal Stress, σ_{1max} (MPa)	Maximum Strain Energy, U_{max} (J)
Regular	2.161	85.22	3.0984e-02
First Order	240.60	87.45	2.0286e-02

TABLE V. NORMALISED RESULTS

Honeycomb	Specific E^* (GPa/kg)	Specific σ_{1max} (MPa/kg)	Specific U_{max} (J/kg)
Regular	0.481	19.00	6.908e-03
First Order	59.85	21.75	5.046e-03

C. Out-of-Plane Tension

This section describes the performance of regular and first order honeycomb structures under in-plane tension with similar boundary conditions and applied force.

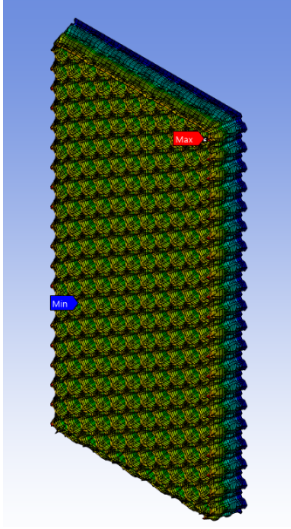


Fig. 18. Directional deformation for regular honeycomb.

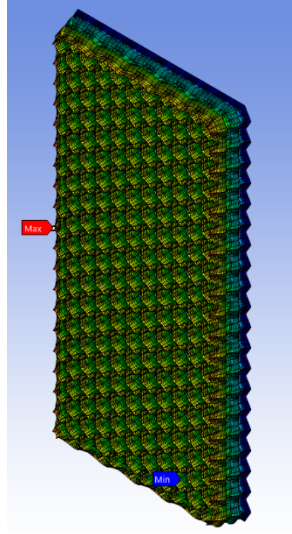


Fig. 19. Directional deformation for first order honeycomb.

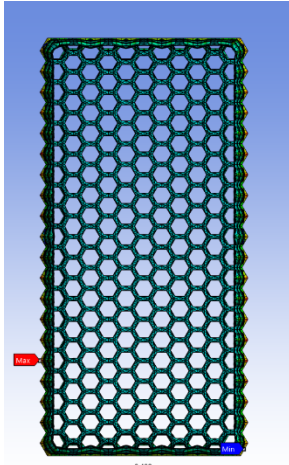


Fig. 20. Maximum principal stress for regular honeycomb.

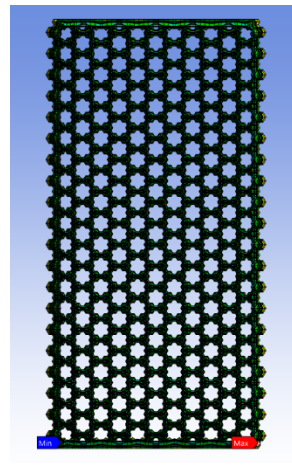


Fig. 21. Maximum principal stress for first order honeycomb

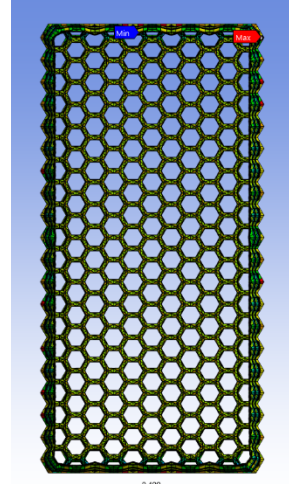


Fig. 22. Maximum strain energy for regular honeycomb.

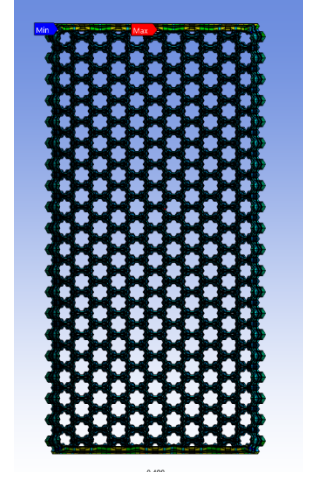


Fig. 23. Maximum strain energy for first order honeycomb.

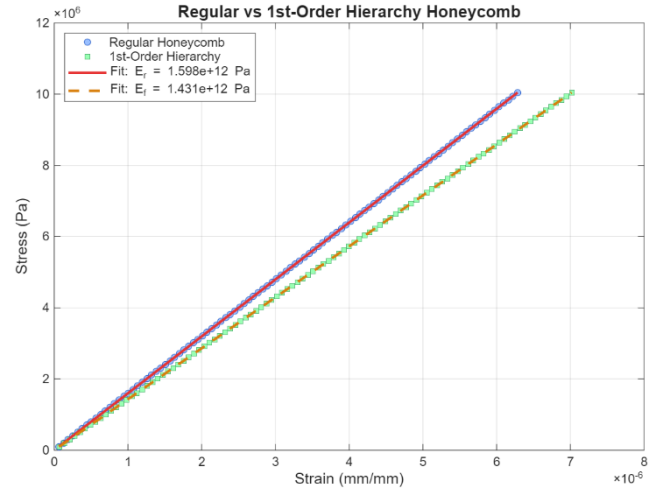


Fig. 24. Effective stiffness plots for regular and first order honeycomb.

TABLE VI. OUT-OF-PLANE TENSION RESULTS

Honeycomb	Effective Stiffness, E^* (GPa)	Maximum Principal Stress, σ_{1max} (MPa)	Maximum Strain Energy, U_{max} (J)
Regular	1598	1.1709	1.0974e-05
First Order	1431	1.4532	1.3274e-05

TABLE VII. NORMALISED RESULTS

Honeycomb	Specific E^* (GPa/kg)	Specific σ_{1max} (MPa/kg)	Specific U_{max} (J/kg)
Regular	356.30	0.261	2.447e-06
First Order	355.97	0.361	3.301e-06

V. DISCUSSION OF RESULTS

A. General Trend

The results compare the mechanical performance of the regular honeycomb and the first-order honeycomb under three loading cases: in-plane tension, in-plane compression, and out-of-plane tension. Overall, the first-order honeycomb generally improves strength and energy absorption, while the regular honeycomb often retains higher or similar stiffness, depending on the loading direction.

B. In-Plane Tension

For in-plane tension in Tables II and III, the regular honeycomb has the higher effective stiffness, with $E^*=597$ GPa compared with 241 GPa for the first-order honeycomb. This means the regular honeycomb is initially more resistant to deformation in tension. However, the first-order honeycomb performs much better in maximum stress and strain energy. Its maximum principal stress is 2780 MPa, compared with 331 MPa for the regular honeycomb, and its maximum strain energy is 2.0286 J compared with 3.1×10^{-2} J. The normalised results show the same pattern: the regular honeycomb has higher specific stiffness, but the first-order honeycomb has much higher specific stress and specific strain energy. Therefore, under in-plane tension, the first-order honeycomb is less stiff but much stronger and absorbs far more energy before failure.

C. In-Plane Compression

For in-plane compression in Tables IV and V, the first-order honeycomb shows a major improvement in stiffness. Its effective stiffness is 240.60 GPa, compared with only 2.161 GPa for the regular honeycomb. This suggests that the first-order geometry is much better at resisting compressive deformation in-plane. The maximum principal stresses are similar, but the first-order honeycomb is slightly higher, at 87.45 MPa compared with 85.22 MPa. However, the regular honeycomb absorbs slightly more strain energy, 3.098×10^{-2} J compared with 2.0286×10^{-2} J. This suggests that in compression, the first-order structure is much stiffer and slightly stronger, while the regular honeycomb may allow more deformation before failure.

D. Out-of-Plane Tension

For out-of-plane tension in Tables VI and VII, the two honeycombs have relatively similar stiffness values. The regular honeycomb is slightly stiffer, with $E^*=1598$ GPa compared with 1431 GPa for the first-order honeycomb. However, the first-order honeycomb again has higher maximum principal stress and higher strain energy. Its maximum principal stress is 1.4532 MPa compared with 1.1709 MPa, and its maximum strain energy is 1.3274×10^{-5} J

compared with 1.0974×10^{-5} J. The normalised results confirm this trend: the specific stiffness values are almost identical, but the first-order honeycomb has better specific strength and specific energy absorption.

E. Takeaway

Overall, the first-order honeycomb provides better strength and energy absorption in both in-plane tension and out-of-plane tension, and it gives a very large improvement in in-plane compressive stiffness. The regular honeycomb is mainly advantageous where higher tensile stiffness is required, especially under in-plane tension and slightly under out-of-plane tension. Therefore, the first-order honeycomb appears more suitable for applications where load-bearing capacity, damage resistance, and energy absorption are important, while the regular honeycomb may be preferable where maximum initial stiffness is the priority.

VI. COMPARISON OF RESULTS IN LITERATURE

A. Effective Stiffness

Although composite material layers were not used in his paper, Ajdari *et al.* [10] concluded that the first-order hierarchy was about 2 times stiffer than regular honeycomb geometries when subjected to axial loading, while also keeping density fixed across all simulations and setting $\gamma_1 = 0.32$. The composite first-order hierarchy honeycomb had 2 orders of magnitude greater stiffness than the default honeycomb geometry when subjected to compression, while only keeping 40% of the stiffness. These results differ most likely due to the composite material's symmetric makeup of carbon fiber and Onyx layers, as their material properties may contribute to an increase or decrease in effective stiffness by orders of magnitude. Additionally, different loading conditions were used which affected the magnitude of results. Nevertheless, all results conclude that the hierarchical honeycomb performs better in axial compression than the default honeycomb.

A first-order hierarchical honeycomb simulated and physically tested for in-plane compression using PLA [11] performed with 1.96 higher stiffness than the default honeycomb geometry. To add, they increased the hierarchy another time which made stiffness 3.03 times greater, further proving that increasing hierarchy increases compressive stiffness.

B. Maximum Principal Stress

Li [12] uses three-point bending test in ABAQUS to analyze the bending performance of aluminum alloy sandwich beams consisting of 6mm wall thickness hierarchical honeycomb structures. According to the literature, hierarchical honeycombs exhibit a bending strength 1.4 times to 1.8 times greater than conventional honeycombs, depending on the scale ratio of the vertex cells. Therefore, the architecture spreads the maximum principal stress across a broader area of the core, which allows for a larger portion of the honeycomb network to participate in energy absorption and load resistance.

However, it is important to note that when these hierarchical honey cores were adjusted to have the same weight parameters as the traditional honeycomb cores by adjusting the cell wall thickness, the sandwich beams almost had the same equivalent bending stiffness. This suggests that while hierarchy improves the peak load-bearing capacity and strength, the stiffness gains are primarily driven by the mass distribution rather than the architectural geometry alone when weight is held constantly.

VII. MANUFACTURING COMPARISON

	Honeycomb	
	<i>Without CF reinforcement</i>	<i>With CF reinforcement</i>
Printing Time (min/g)	10.2	8.5

	First-Order Hierarchy Honeycomb	
	<i>Without CF reinforcement</i>	<i>With CF reinforcement</i>
Printing Time (min/g)	13.9	11.1

Without carbon fiber reinforcements, the first-order hierarchy honeycomb takes 36.89% longer to print per gram than the normal honeycomb. With carbon fiber reinforcements, it takes 30.41% longer.

VIII. LIMITATIONS

1. Physical testing of the printed specimens was ultimately precluded by constraints on the maximum specimen dimensions permitted by the tensions test in Hesse Hall, which restricts specimen sizes to a maximum of 12.7 mm x 381 mm x 25.4 mm. Due to our cell wall thickness being 3.75mm, the maximum allowable specimen width would only allow for 2-unit cells. At this scale, the edge and boundary effects would dominate the results, rendering our experimental data inadequate to compare with our simulation results.

2. We limited the scope of this study to first-order hierarchies because progressing to a second-order hierarchy would require cell sizes that exceed the limits of the Markforged FDM printer. Furthermore, the marginal mechanical improvements associated with higher-order hierarchies are often outweighed by the increase in complex geometry and manufacturing demands, making them impractical for this specific application.

3. We assumed the Markforged Onyx to be isotropic, however, because FDM printing deposits material layer by layer, the final part should be anisotropic. Therefore, our FEA might overestimate out-of-plane tension because it does not account for layer delamination.

IX. CONCLUSION

The first-order hierarchical honeycomb geometry has both mechanical advantages and disadvantages compared to the default honeycomb geometry. Users should be mindful about the placement of the honeycomb structure when utilizing the first-order hierarchy of geometry, as it only performs better in in-plane compression. Failure can be catastrophic when placed in a position where it's being pulled apart.

A comparison between first order and standard honeycomb structures reveals that while the regular honeycomb is stiffer in tension, the first-order hierarchy allows for more strain and deformation before failure. In contrast, the first-order hierarchy is significantly stiffer in compression, leading to a very brittle failure mode. The two geometries exhibit more similar behavior in out-of-plane tension, and may slightly reduce stiffness by introducing additional internal ligaments and stress redistribution rather than direct axial resistance.

The first-order hierarchy honeycomb structure outperforms its default honeycomb structure in weight, as it is 10.4% lighter. This weight difference could provide benefits for products that perform better with lightweight structures, such as automobiles, planes, rockets, and more, as fuel efficiency increases as weight decreases. However, regardless of the honeycomb structure made up of composite materials, the 3D printing time increases by about 35% when printing the first-order hierarchy honeycomb. Even though its lightweight quality causes the geometry itself to cost less to print, the first-order honeycomb can increase production costs due to needed a longer time to manufacture.

While this study provides valuable insight into hierarchical composites, limitations offer opportunities for future research. The test geometry constraints prevented us from validating our simulation results with physical 3D printed test specimens. Furthermore, the accuracy of the simulations could be improved by refining the fiber directions as opposed to our study's method of placing fibers uniformly in one direction. In future work, since ACP (Pre) treats plies with ideal bonding, more investigation into delamination could help provide better understanding of honeycomb hierarchical geometries.

Ultimately, this research provides a clearer understanding of how hierarchical ordering influences the mechanical properties of composite honeycombs, offering engineers and the industry more options for lightweight structural design.

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